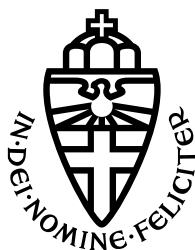


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The Prosciutto Question

Bachelor Thesis in Mathematics

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Chapter 1

Preliminaries

1.1 General knowledge and notation

We begin by recalling and defining some general tools that play an important role throughout this thesis. The first few concepts are often used to translate local properties into global results. This method plays a fundamental part in our later discussions about the formulation and analysis of weak forms and solutions to PDEs.

Let X be a topological space. A function $\phi : X \rightarrow \mathbb{R}$ is said to have *compact support* if the set

$$\text{supp}(\phi) := \overline{\{x \in X \mid \phi(x) \neq 0\}}$$

is a compact subset of X . For any open $\Omega \subseteq \mathbb{R}^d$, the space of all smooth functions with compact support in Ω is denoted by $C_c^\infty(\Omega)$.

An important property of these functions is that they must be 0 on the boundary of Ω . In the particular context of PDE analysis, they are often referred to as *test functions*, but a more general term is *variations*. They have a wide range of applications, discussed in the field of the Calculus of Variations. A large portion of its theory is built around the following result, hence its name.

Lemma 1.1 (Fundamental Lemma of the Calculus of Variations)

Let $\Omega \subseteq \mathbb{R}^d$ be open and $f \in L^1_{\text{loc}}(\Omega)$. If

$$\int_{\Omega} f \phi = 0$$

holds for all $\phi \in C_c^\infty(\Omega)$, then $f = 0$ almost everywhere in Ω .

Multiplying with a test function and integrating is sometimes referred to as *testing against* a test function $\phi \in C_c^\infty(\Omega)$.

Definition 1.2

When we are differentiating functions in C^k defined in a subset of $\mathbb{R}^d$, we sometimes use the abbreviated notation

$$D^\alpha := \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}},$$

where $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$ is called a *multi-index* and $|\alpha| := \alpha_1 + \dots + \alpha_d \leq k$.

Definition 1.3

Let $\Omega \subseteq \mathbb{R}^d$ be open. A function $f : \Omega \rightarrow \mathbb{R}$ is said to be *locally integrable* if

$$\int_K |f| < \infty$$

for all compact subsets K of Ω . The space of all locally integrable functions is denoted by $L^1_{\text{loc}}(\Omega)$, where we identify two functions with each other if they agree almost everywhere in Ω .

Put this before FLCV

An important fact that should be noted is that every function that is square-integrable is also locally integrable. That is, $L^2(\Omega) \subseteq L^1_{\text{loc}}(\Omega)$. For the sake of completeness, we will sometimes state definitions or results with the hypothesis that a function is locally integrable, only to use them on functions that are square-integrable.

Definition 1.4

Let $a < b$. A function $f : [a, b] \rightarrow \mathbb{R}$ is called *absolutely continuous* if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for any finite collection of subintervals $\{(x_k, y_k)\}_{k=1}^N$ of $[a, b]$,

$$\sum_{k=1}^N (y_k - x_k) < \delta \implies \sum_{k=1}^N |f(y_k) - f(x_k)| < \varepsilon.$$

Proposition 1.5

A function $f : [a, b] \rightarrow \mathbb{R}$ is absolutely continuous if and only if f has derivative f' almost everywhere, f' is integrable, and

$$f(x) = f(a) + \int_a^x f'(t) dt$$

for every $x \in [a, b]$.

Definition 1.6

Let n be a positive integer, I an interval, $x : I \rightarrow \mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}^n$ for some set $S \subseteq \mathbb{R} \times \mathbb{R}^n$. Then x is called a *solution* of

$$x'(t) = f(t, x(t)) \quad (1.1)$$

in the sense of Carathéodory on I if x is absolutely continuous, $(t, x(t)) \in S$ for every $t \in I$ and Equation (1.1) is satisfied for almost every $t \in I$.

Theorem 1.7 (Carathéodory's existence theorem)

Let n be a positive integer, $a, b > 0$ and $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$. Define

$$R := \{(t, x) \in \mathbb{R} \times \mathbb{R}^n \mid |t - t_0| \leq a \text{ and } \|x - x_0\| \leq b\}$$

and let $f : R \rightarrow \mathbb{R}^n$ be continuous in x for every fixed t , measurable in t for every fixed x and bounded by some time-dependent integrable function m ,

$$\|f(t, x)\| \leq m(t)$$

for all $(t, x) \in R$. Then the differential equation

$$x'(t) = f(t, x(t))$$

has a solution in the sense of Carathéodory on the interval $|t - t_0| \leq \beta$ for some $\beta > 0$, satisfying $x(0) = x_0$.

Theorem 1.8 (Nagumo's invariance theorem)

Let $M \subseteq \mathbb{R}^n$ be closed and $J = [a, b]$ some real interval. Assume that $f : D \subseteq J \times M \rightarrow \mathbb{R}^n$ satisfies the tangent condition

$$(\nu(x), f(t, x)) \leq 0 \quad \text{for } t \in [\xi, \xi + a] \text{ and } x \in \partial M,$$

where $\nu(x)$ is the outward normal vector to M at x , as well as a condition of Lipschitz type

$$(x_1 - x_2, f(t, x_1) - f(t, x_2)) \leq L \|x_1 - x_2\|^2.$$

Then M is flow-invariant with respect to the differential equation $x' = f(t, x)$.

more reg-
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for f ?

1.2 Weak derivatives and Sobolev spaces

Before we talk about weak solutions to PDEs, we need to define spaces of functions that are potential candidates for solutions. We derive this space by weakening the conditions of differentiable

functions.

Consider an open and bounded domain $\Omega \subseteq \mathbb{R}^d$. Then for any continuously differentiable function $u \in C^1(\Omega)$ and test function $\phi \in C_c^\infty(\Omega)$, we have

$$\int_{\Omega} u_{x_i} \phi = \int_{\partial\Omega} \frac{\partial u}{\partial \nu} \phi \, dS - \int_{\Omega} u \phi_{x_i} = - \int_{\Omega} u \phi_{x_i}, \quad (1.2)$$

since ϕ is 0 on the boundary. Consequentially, for any function $u \in C^k(\Omega)$ and multi-index $\alpha \in \mathbb{N}^d$ with $|\alpha| \leq k$, we have

$$\int_{\Omega} D^\alpha u \phi = (-1)^{|\alpha|} \int_{\Omega} u D^\alpha \phi, \quad (1.3)$$

using the fact that $D^\beta \phi$ has compact support as well, for every $\beta \in \mathbb{N}^d$ with $|\beta| \leq k$. Note that for this last expression to make sense, the function $D^\alpha u$ does not necessarily have to be continuous. In fact, we can completely remove the hypothesis that u is k times continuously differentiable, and define the *weak derivative* of u as a function that behaves in a similar way as Equation (1.3) when tested against a function $\phi \in C_c^\infty(\Omega)$.

Definition 1.9

Let $u, v \in L^1_{\text{loc}}(\Omega)$ and $\alpha \in \mathbb{N}^d$. We say that v is the α -th *weak derivative* of u if

$$\int_{\Omega} v \phi = (-1)^{|\alpha|} \int_{\Omega} u D^\alpha \phi \quad (1.4)$$

for every test function $\phi \in C_c^\infty(\Omega)$.

We often use the same notation for weak derivatives as we do for regular derivatives. When $D^\alpha u$ is used, we should carefully think about which derivative is meant here. This notion extends to any differential operator that can classically be written in terms of partial derivatives, like the gradient ∇ or Laplacian $\Delta := \frac{\partial^2}{\partial x_1^2} + \cdots + \frac{\partial^2}{\partial x_d^2}$. To be able to talk about *the* α -th weak derivative, we show that weak derivatives are unique.

Proposition 1.10

If the α -th weak derivative of u exists, it is unique up to a set of measure zero.

The proof of this proposition is a good example of how the Fundamental Lemma of the Calculus of Variations can be used.

Proof. Suppose that v and w are two α -th weak derivatives of u . Then for every $\phi \in C_c^\infty(\Omega)$,

$$\int_{\Omega} v \phi = (-1)^{|\alpha|} \int_{\Omega} u D^\alpha \phi = \int_{\Omega} w \phi,$$

which gives

$$\int_{\Omega} (v - w)\phi = 0$$

for every $\phi \in C_c^\infty(\Omega)$. We can now use the Fundamental Lemma of the Calculus of Variations (**FLCV**) to see that $v - w$ is 0 almost everywhere, which means that $v = w$ almost everywhere, completing the proof. $\square$

With this new notion of weak differentiability, we can define new spaces of regularity. These spaces are fundamental to the analysis of PDEs and will be used a lot in this thesis.

Definition 1.11

For $k \in \mathbb{N}$ and $p \in [1, \infty]$, the *Sobolev space of order k* , denoted $W^{k,p}(\Omega)$, is the space of functions for which the α -th weak derivative exists and is in $L^p(\Omega)$, for all $|\alpha| \leq k$:

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega) \mid D^\alpha u \in L^p(\Omega), |\alpha| \leq k\}.$$

When equipped with the norm

$$\|u\|_{W^{k,p}(\Omega)} := \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p},$$

the space becomes complete, and thus forms a Banach space.

Recall that $L^p(\Omega)$ is a Banach space for every $p \in [1, \infty]$, but for $p = 2$ it also forms a Hilbert space. We can say this for Sobolev spaces as well. We write $H^k(\Omega) := W^{k,2}(\Omega)$ and equip it with the inner product

$$(u, v)_{H^k(\Omega)} := \sum_{|\alpha| \leq k} (D^\alpha u, D^\alpha v)_{L^2(\Omega)},$$

forming a Hilbert space.

A useful subset of $W^{k,p}(\Omega)$ is $W_0^{k,p}(\Omega)$, which is defined as the closure of $C_c^\infty(\Omega)$ as a subset of $W^{k,p}(\Omega)$. For the case of $k = 2$, we write $H_0^k(\Omega) := W_0^{k,2}(\Omega)$.

Lastly, we define and $H^{-1}(\Omega)$ as the dual space of $H_0^1(\Omega)$, and denote the pairing between these two spaces as $\langle \cdot, \cdot \rangle$, i.e. $\langle f, u \rangle := f(u)$.

When $\partial\Omega$ is sufficiently smooth, we have a more intuitive way to interpret the space $H^1(\Omega)$. In fact, we can describe $H_0^1(\Omega)$ as the subset of $H^1(\Omega)$ containing all functions that vanish on the boundary. This interpretation has an important caveat. By definition, $H^1(\Omega)$ is a subset of $L^2(\Omega)$, so the functions that make up this Sobolev space are only defined up to a set of measure zero. The problem lies in the fact that the boundary $\partial\Omega$ is a set of measure zero, so we can generally not say anything about the values that are attained on it yet. We can, however, consider the case where we have a continuously differentiable function $u \in C^1(\overline{\Omega})$, for which the values on $\partial\Omega$ are certainly

defined. We can also see u as a function in $H^1(\Omega)$, and define $v|_{\partial\Omega} := u|_{\partial\Omega}$ for every $v \in H^1(\Omega)$ that is equal to u almost everywhere in Ω . To further develop this idea, we first state the following property.

Proposition 1.12 (Approximation by smooth functions)

If the boundary $\partial\Omega$ is Lipschitz and $1 \leq p < \infty$, the set of all smooth functions $C^\infty(\overline{\Omega})$ is dense in $W^{k,p}(\Omega)$.

More precisely, for any function $u \in W^{k,p}(\Omega)$, there is a sequence of smooth functions defined on the closure of Ω , $(u_n)_{n \in \mathbb{N}} \subseteq C^\infty(\overline{\Omega})$, such that

$$\|u - u_n\|_{W^{k,p}(\Omega)} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

This allows us to extend our knowledge of functions defined on the boundary to functions in $W^{k,p}(\Omega)$ in general.

Theorem 1.13 (Trace Theorem)

If the boundary $\partial\Omega$ is Lipschitz, there is a bounded linear operator $\gamma : W^{1,p}(\Omega) \rightarrow L^p(\partial\Omega)$ such that

- (i) $\gamma u = u|_{\partial\Omega}$ for any $u \in W^{1,p}(\Omega) \cap C(\overline{\Omega})$ and
- (ii) $\|\gamma u\|_{L^2(\partial\Omega)} \leq C \|u\|_{W^{1,p}(\Omega)}$,

where the constant $C > 0$ only depends on p and Ω .

A full proof of this theorem can be found in [Eva10], but we will describe the general outline here.

Proof. As we previously discussed, we can start by defining $\gamma u := u|_{\partial\Omega}$ for any $u \in C^1(\overline{\Omega})$. The first step in extending this definition is to use that the boundary is Lipschitz to locally flatten it and prove item (ii) for $u \in C^1(\overline{\Omega})$ in a neighbourhood of some point $x \in \partial\Omega$. We can then use the fact that Ω is bounded to see that $\partial\Omega$ is compact, which allows us to get a finite amount of points $x_1, \dots, x_n \in \partial\Omega$ with corresponding constants $C_1, \dots, C_n$, combining them to conclude item (ii) for the entirety of $\partial\Omega$.

For the general case of $u \in W^{1,p}(\Omega)$, we can use Proposition 1.12 to get an approximation for u of smooth functions $(u_n)_{n \in \mathbb{N}} \subseteq C^\infty(\overline{\Omega})$. The first part of the proof then implies that

$$\|\gamma u_n - \gamma u_m\|_{L^2(\partial\Omega)} = \|\gamma(u_n - u_m)\|_{L^2(\partial\Omega)} \leq C \|u_n - u_m\|_{W^{1,p}(\Omega)},$$

which means that $(\gamma u_n)_{n \in \mathbb{N}}$ is a Cauchy sequence. We have used that γ is linear on $C^1(\overline{\Omega})$, since it simply acts as a restriction there. Using the fact that $L^2(\partial\Omega)$ is a Hilbert space, we can define

$$\gamma u := \lim_{n \rightarrow \infty} \gamma u_n = \lim_{n \rightarrow \infty} u_n|_{\partial\Omega},$$

which satisfies the same bound.

The proof is completed by noting that if $u \in W^{1,p}(\Omega) \cap C(\bar{\Omega})$, then there are smooth functions $(u_n)_{n \in \mathbb{N}} \subseteq C^\infty(\bar{\Omega})$ converging to u uniformly on $\bar{\Omega}$, so $\gamma u = u|_{\partial\Omega}$. $\square$

Definition 1.14

This operator $\gamma : W^{1,p}(\Omega) \rightarrow L^p(\partial\Omega)$ is referred to as the *trace operator*. Then γu is called the *trace* of u , and is sometimes denoted by $u|_{\partial\Omega}$.

1.3 Weak time derivatives

In the field of PDEs, the requirements on time regularity and space regularity are often different. Let for example

Let X be a Banach space and $T > 0$. A function $f : [0, T] \rightarrow X$ is called *strongly measurable* if there is a sequence of simple functions $(f_n)_{n \in \mathbb{N}}$ that converges pointwise to f at almost every $t \in [0, T]$. Now for $p \in [1, \infty]$, the space $L^p(0, T; X)$ consists of all strongly measurable functions u such that

$$\|u\|_{L^p(0,T;X)} := \left(\int_0^T \|u(t)\|_X^p dt \right)^{1/p} < \infty$$

for $p \in [1, \infty)$, and

$$\|u\|_{L^\infty(0,T;X)} := \operatorname{ess\,sup}_{t \in [0,T]} \|u(t)\|_X < \infty.$$

Lastly, $C([0, T]; X)$ denotes the space of all continuous functions $u : [0, T] \rightarrow X$ with

$$\|u\|_{C([0,T];X)} := \max_{t \in [0,T]} \|u(t)\|_X < \infty.$$

We can take X to be, for instance, $L^2(\Omega)$. Then formally, $C([0, T]; L^2(\Omega))$ consists of continuous functions u from the interval time $[0, T]$ to $L^2(\Omega)$. For a more intuitive interpretation, you might feel inclined to interpret them as the corresponding functions $\tilde{u}$ from $[0, T] \times \Omega$ to $\mathbb{R}$, given by $\tilde{u}(t, x) = u(t)(x)$. We do need to be careful when we use this interpretation, though, as the notation might make it seem like these functions $\tilde{u}$ are continuous in time, while this is generally not the case. A more accurate way to think about this, is that at every time t , the set of points $x \in \Omega$ where $\tilde{u}$ has a discontinuity in time has measure zero.

Caratheodory
?

1.4 Variational forms

Using the same method, we can weaken the requirements for solutions to a PDE. Consider, for instance, the heat equation on $\Omega_T := (0, T) \times \Omega$ with homogeneous Dirichlet boundary conditions. For a function u to be a classical solution, we require u to be of class $C^{1,2}(\Omega_T) \cap C^0(\bar{\Omega}_T)$, i.e. continuously differentiable with respect to t , twice continuously differentiable with respect to x and

continuously extendable to $\overline{\Omega_T}$. We can multiply the heat equation by a test function $\phi \in C_c^\infty(\Omega)$ and integrate on Ω to see

rewrite

$$\int_{\Omega} u_t \phi = \int_{\Omega} \Delta u \phi = \int_{\partial\Omega} \frac{\partial u}{\partial \nu} \phi \, dS - \int_{\Omega} \nabla u \cdot \nabla \phi = - \int_{\Omega} \nabla u \cdot \nabla \phi. \quad (1.5)$$

We can also write this as

$$\langle u_t, \phi \rangle = - \langle \nabla u, \nabla \phi \rangle, \quad (1.6)$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product on both $L^2(\Omega)$ and $L^2(\Omega; \mathbb{R}^m)$.

Chapter 2

Turing patterns

Chapter 3

A bulk-surface model

In this chapter, we will discuss a bulk-surface model that is motivated by the interactions of ligands and receptors. We will give a classical proof of existence and uniqueness of weak solutions to this model, as well as discuss some of the qualitative properties of such solutions. The proof will rely heavily on the definitions and results introduced in [chapter 1](#), and will be based on an application of the Banach fixed-point theorem.

3.1 Biological motivation

Consider an open and bounded set $U \subseteq \mathbb{R}^d$ with C^1 boundary and $T > 0$. This can be interpreted as the microscopic part of the body we are investigating. We identify the cells in this part of the body by an open set $C \subseteq U$, also with C^1 boundary, and the extracellular space by $\Omega := U \setminus \overline{C}$. We require that the closure $\overline{C}$ of C is also completely contained by U , i.e. no cells should cross the boundary of our domain. In particular, this implies that the boundary $\partial\Omega$ of the extracellular space can be written as the disjoint union of the boundary of our region of investigation ∂U and the cell membranes ∂C , which we denote by Γ_e and Γ , respectively. We use the function $u : [0, T] \times \Gamma \rightarrow \mathbb{R}$ to represent the concentration of free receptors at a given time and place on a cell membrane. Similarly, $u_b : [0, T] \times \Gamma \rightarrow \mathbb{R}$ represents the concentration of bounded receptors, and $v : [0, T] \times \overline{\Omega} \rightarrow \mathbb{R}$ represents the concentration of ligands.

actual bio-
logical terms

The ligands can move freely through the extracellular space. This can be modelled by the standard diffusion (or heat) equation. For the sake of simplification, we assume the individual receptors will be fixed in their position on the cell membrane.

figure

When a ligand binds to a free receptor, the concentration of free receptors and ligands decrease, and the concentration of bounded receptors at that position increases accordingly. The rate of this reaction is proportional to $u^2 v$. After some time, the bounded receptor has done its task, passing the signal through to the interior of the cell. It releases the ligand and is now available to bind to a new one again. The concentration of bounded receptors decrease, and the concentration of free receptors and ligands increase accordingly. The rate of this dissociation is proportional to the concentration of bounded receptors u_b .

or uv

The last two things that happen constantly are production and decay. To keep intercellular communication possible, cells produce ligands and receptors that enter or model through the cell membrane. We assume this production mainly happens at a constant rate, but is also upregulated

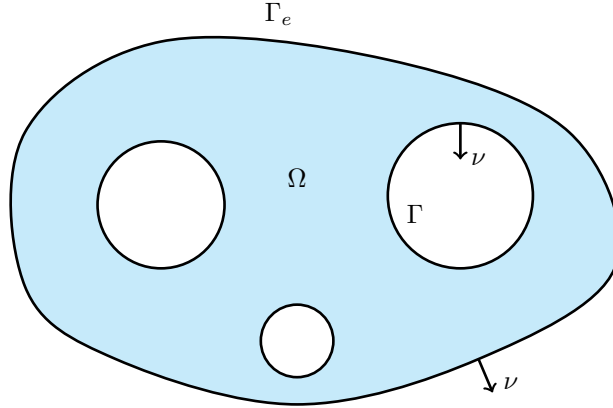


Figure 3.1: A sketch of a possible domain in the case that $d = 2$. The blue background represents Ω and the two kinds of boundaries are indicated, along with the outward unit normal vector ν .

by the concentration of bounded receptors. The ligands and receptors decay at a rate proportional to their respective concentrations.

We can describe this behaviour with a system of coupled differential equations. Denote the association and dissociation coefficients by $k_{\text{on}}, k_{\text{off}} > 0$, respectively, the production constants by $p_u, p_v > 0$, the upregulation coefficient by $\alpha > 0$, the decay coefficients by $d_u, d_{u_b} > 0$ and the diffusion coefficient by $D > 0$. Furthermore, we denote the outward unit normal vector on $\partial\Omega$ by ν . Then the interactions between ligands and receptors are described by

$$f(u, u_b, v) := k_{\text{on}}u^2v - k_{\text{off}}u_b,$$

and the production and decay of u, u_b and v are described by

$$p(u, u_b) := p_u + \alpha u_b - d_u u, \quad q(u_b) := -d_{u_b} u_b \quad \text{and} \quad r := p_v,$$

respectively. The concentrations of receptors can then be described by the system of ODEs given by

$$\begin{aligned} u_t &= -2f(u, u_b, v) + p(u, u_b) & \text{on } (0, T) \times \Gamma, \\ u_{b,t} &= f(u, u_b, v) + q(u_b) & \text{on } (0, T) \times \Gamma. \end{aligned} \tag{3.1}$$

Note that even though they are given by only two equations, this is actually a system of uncountably many ODEs. For every boundary point $x \in \Gamma$, we have one ODE for the function $t \mapsto u(t, x)$ and one for the function $t \mapsto u_b(t, x)$.

Since the dynamics of the ligands include diffusion, they can be described by the PDEs given by

$$\begin{aligned} v_t &= D\Delta v - d_v v & \text{in } (0, T) \times \Omega, \\ D \frac{\partial v}{\partial \nu} &= -f(u, u_b, v) + r(v) & \text{on } [0, T] \times \Gamma, \\ D \frac{\partial v}{\partial \nu} &= 0 & \text{on } [0, T] \times \Gamma_e. \end{aligned} \tag{3.2}$$

Additionally, these concentrations can be required to adhere to some initial conditions. That is, they should satisfy

$$\begin{aligned} u(0, \cdot) &= u_0 && \text{on } \Gamma, \\ u_b(0, \cdot) &= u_{b,0} && \text{on } \Gamma, \\ v(0, \cdot) &= v_0 && \text{on } \Omega, \end{aligned} \tag{3.3}$$

for some given $u_0, u_{b,0} : \Gamma \rightarrow \mathbb{R}$ and $v_0 : \Omega \rightarrow \mathbb{R}$.

The remainder of this chapter will be dedicated to studying the system of Equations (3.1) to (3.3).

3.2 Weak formulation

Before we start proving the existence of solutions to this model, we must discuss what kind of solutions we are looking for. Specifically, we will derive the weak formulations of Equations (3.1) to (3.3) and discuss the function spaces that are involved.

We first consider the PDE, proceeding similarly to section 1.4. To state the weak formulation, we assume $\tilde{u}$, $\tilde{u}_b$ and $\tilde{v}$ are smooth solutions to Equations (3.1) to (3.3) and note that we can also interpret them as the corresponding functions $u \in \mathcal{X}$, $u_b \in \mathcal{X}$ and $v \in \mathcal{Y}$ given by

$$u(t) = \tilde{u}(t, \cdot), \quad u_b(t) = \tilde{u}_b(t, \cdot) \quad \text{and} \quad v(t) = \tilde{v}(t, \cdot).$$

We denote the weak time derivative of v by v' , multiply it by a test function $\phi \in C_c^\infty(\Omega)$ and integrate on Ω for fixed t , giving

$$\begin{aligned} \int_{\Omega} v'(t) \phi &= D \int_{\Omega} \Delta v(t) \phi - d_v \int_{\Omega} v(t) \phi \\ &= -D \int_{\Omega} \nabla v(t) \cdot \nabla \phi + D \int_{\partial\Omega} \frac{\partial v(t)}{\partial \nu} \phi \, dS - d_v \int_{\Omega} v(t) \phi \\ &= -D \int_{\Omega} \nabla v(t) \cdot \nabla \phi + D \int_{\Gamma} \frac{\partial v(t)}{\partial \nu} \phi \, dS + D \int_{\Gamma_e} \frac{\partial v(t)}{\partial \nu} \phi \, dS - d_v \int_{\Omega} v(t) \phi \\ &= -D \int_{\Omega} \nabla v(t) \cdot \nabla \phi - \int_{\Gamma} f(u(t), u_b(t), v(t)) \phi \, dS + \int_{\Gamma} r(v(t)) \phi \, dS - d_v \int_{\Omega} v(t) \phi, \end{aligned}$$

which we can write using inner product notation as

$$(v'(t), \phi)_{\Omega} + d_v (v(t), \phi)_{\Omega} = -D (\nabla v(t), \nabla \phi)_{\Omega} - (f(u(t), u_b(t), v(t)), \phi)_{\Gamma} + (r(v(t)), \phi)_{\Gamma}. \tag{3.4}$$

Since we used integration with respect to space in this computation, this expression only depends on time. A weak solution of Equation (3.2) is therefore required to satisfy Equation (3.4) at almost every $t \in [0, T]$.

For the ODEs, we want to use Theorem 1.7, i.e. we will be looking for solutions of the system in the sense of Carathéodory on $[0, T]$. This weakened problem only depends on time, so we fix $x \in \Gamma$ and look for an absolutely continuous function $w := (u(\cdot, x), u_b(\cdot, x))$ such that

$$w'(t) = F(w(t), t) \quad (3.5)$$

almost everywhere, where $F : \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is given by

$$F(t, (u, u_b)) = \begin{pmatrix} -2f(u, u_b, \gamma v(t, x)) + p(u) \\ f(u, u_b, \gamma v(t, x)) + q(u_b) \end{pmatrix} = \begin{pmatrix} -2k_{on}u^2\gamma v(t, x) + 2k_{off}u_b + p_u + \alpha u_b - d_u u \\ k_{on}u^2\gamma v(t, x) - k_{off}u_b - d_{u_b}u_b \end{pmatrix}, \quad (3.6)$$

where γv denotes the trace of v as discussed in section 1.2. Note that this definition of F does not take into account that originally, the existence of v is problem we need to solve simultaneously. We will justify this in the next section, when we separate the problems for the ODEs and the PDE and see that we can assume v to be given. Also note that this definition depends on the fact that x is assumed to be fixed. Thus for any $x \in \Gamma$, this defines a different problem.

We can now combine these two formulations and start discussing existence and uniqueness properties.

define $\mathcal{X}$ and $\mathcal{Y}$

Definition 3.1

The functions $u, u_b \in \mathcal{X}$ and $v \in \mathcal{Y}$ are *weak solutions* to Equations (3.1) to (3.3) if v satisfies Equation (3.4) for almost every $t \in [0, T]$ and

$$w(t) = (u(t, x), u_b(t, x))$$

satisfies Equation (3.5) for almost every $t \in [0, T]$ and $x \in \Gamma$.

correct?

3.3 Existence and uniqueness

Our goal in this section is to prove the following theorem.

Theorem 3.2 (Existence and uniqueness of weak solutions)

Let $u_0, u_{b,0} \in L^2(\partial\Omega)$ and $v_0 \in L^2(\Omega)$ be non-negative initial conditions. Then there exist unique weak solutions $u, u_b \in \mathcal{X}$ and $v \in \mathcal{Y}$ to Equations (3.1) to (3.3). Moreover, these solutions are also non-negative.

As previously mentioned, the proof will be based on an application of the Banach fixed-point theorem. If you have followed a course on ODE theory, chances are that you have seen the Picard-Lindelöf Theorem and its proof. The proof for our model follows the same general idea, but requires some extra steps to handle our situation of a coupled system. We “freeze” the terms in the ODEs that depend on v so we can use the standard existence and uniqueness properties we know about such systems to get solutions u and u_b to this alternative problem. The PDE gets a similar treatment, producing solutions v of some alternative PDE, after which we can “update” the frozen terms in the ODE. This produces an operator $\mathcal{T}$ that maps any $v \in \mathcal{Y}$ to another $\tilde{v} \in \mathcal{Y}$. If we can

then show that this operator is a contraction, the Banach fixed-point theorem will guarantee the existence and uniqueness of a fixed point $v^* \in \mathcal{V}$, which will be our solution to the PDE.

Let us first concretely demonstrate what we mean by “freezing” the terms that depend on v . Fix $v^* \in \mathcal{V}$ and consider the modified system of ODEs given by

$$\begin{aligned} u_t &= -f(u, u_b, \gamma v^*) + p(u) & \text{on } (0, T) \times \Gamma, \\ u_{b,t} &= f(u, u_b, \gamma v^*) + q(u_b) & \text{on } (0, T) \times \Gamma. \end{aligned} \quad (3.7)$$

weak form?

These ODEs are now *decoupled* from the PDE, so we can freely talk about the existence and uniqueness of solutions to this system without simultaneously having to worry about v satisfying the PDE.

Lemma 3.3 (Existence and uniqueness for the decoupled ODEs)

Let $u_0, u_{b,0} \in L^2(\partial\Omega)$ be non-negative initial conditions and fix $v^* \in \mathcal{V}$. Then there exist unique weak solutions $u, u_b \in \mathcal{X}$ to Equation (3.7) satisfying these initial conditions. Moreover, these solutions are also non-negative and bounded by a constant depending on T .

Proof. Fix x and define w and $F : \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ as in Equations (3.5) and (3.6), where v is replaced by γv^* . Since the components of F are polynomial in w , it is certainly continuous in w . Additionally, by regularity of γv^* , we know that F is measurable in t . This also implies that for any compact subset of $\mathbb{R} \times \mathbb{R}^2$, we can bound F with some time-dependent integrable function m . In particular, this subset can be taken to be a rectangle¹ as in the hypothesis of Theorem 1.7, so let R be such a rectangle around the initial time $t = 0$ and position $(u_0, u_{b,0})$. The hypothesis of Theorem 1.7 is then fully satisfied, giving us a local solution $w : [0, \beta] \rightarrow \mathbb{R}^2$ for some $\beta > 0$, with $w(0) = (u_0, u_{b,0})$.

time regularity of trace

We show that this local solution is unique. Suppose we have two such local solutions, w_1 and w_2 . Since these solutions are absolutely continuous, they are bounded on $[0, \beta]$, so we can take a compact set $K \subseteq \mathbb{R}^2$ that contains both their trajectories. Again, F is polynomial in w and measurable in time, so it satisfies a Lipschitz-type condition

right?

$$\|F(t, \bar{w}) - F(t, \tilde{w})\| \leq L_K(t) \|\bar{w} - \tilde{w}\|$$

for all $t \in [0, \beta]$ and $\bar{w}, \tilde{w} \in K$, where $L_K : [0, \beta] \rightarrow \mathbb{R}$ is an integrable function. Now the distance between the two solutions satisfies

$$\begin{aligned} \|w_1(t) - w_2(t)\| &= \left\| (u_0, u_{b,0}) + \int_0^t f(t, w_1(t)) dt - (u_0, u_{b,0}) - \int_0^t f(t, w_2(t)) dt \right\| \\ &\leq \int_0^t \|f(t, w_1(t)) - f(t, w_2(t))\| dt \\ &\leq \int_0^t L_K(t) \|w_1(t) - w_2(t)\| dt. \end{aligned}$$

for every $t \in [0, \beta]$. Using Gronwall’s inequality, we see that this implies that $\|w_1(t) - w_2(t)\| = 0$ for all $t \in [0, \beta]$, so $w_1 = w_2$.

¹On first sight, it can seem strange to call this type of set a rectangle instead of a cylinder. This name comes from the fact that in our main source [CL84], the theorem is proven for the case of $n = 1$.

To show that the components u and u_b of this solution w are non-negative, we use Theorem 1.8. □

 Lipschitz
condition?

Using Lemma 3.3, we define an operator $\mathcal{T}_1 : \mathcal{Y} \rightarrow \mathcal{X}^2$ that maps any function $v^* \in \mathcal{Y}$ to a tuple $(u, u_b) \in \mathcal{X}^2$ of solutions to Equation (3.7).

We can use a similar method for the PDE in Equation (3.2). Fix $(u^*, u_b^*) \in \mathcal{X}^2$ and consider the modified weak formulation given by

$$\begin{aligned} (v_t, \phi)_{L^2(\Omega)} &= -(f(u^*, u_b^*, v), \phi)_{L^2(\Gamma)} + (r(v), \phi)_{L^2(\Gamma)} - D(\nabla v, \nabla \phi)_{L^2(\Omega; \mathbb{R}^d)} \\ &= (g(v), \phi)_{L^2(\Gamma)} - D(\nabla v, \nabla \phi)_{L^2(\Omega; \mathbb{R}^d)}, \end{aligned} \quad (3.8)$$

where $g(v) := -f(u^*, u_b^*, v) + r(v)$. Note that we can only make this definition because u^* and u_b^* are fixed. We have now decoupled the PDE from the ODEs, and we can prove a result that is nearly identical to Lemma 3.3.

Lemma 3.4 (Existence and uniqueness for the decoupled PDE)

Let $v_0 \in L^2(\Omega)$ be non-negative initial conditions and fix $(u^*, u_b^*) \in \mathcal{X}^2$. Then there exists a unique weak solution $v \in \mathcal{Y}$ to Equation (3.8) satisfying these initial conditions. Moreover, this solution is also non-negative and bounded by a constant depending on T .

Proof. Now let v be such a local solution. To show that v is non-negative, we estimate the negative part of v . Let v_+ and v_- be defined as $\max(0, v)$ and $\min(0, v)$ respectively, such that $v = v_+ + v_-$. Then we also have $\frac{\partial v}{\partial t} v_- = \frac{\partial v_-}{\partial t} v_-$, as well as $\nabla v \cdot \nabla v_- = \|\nabla v_-\|^2$, so

 Existence
and Uniqueness

$$\begin{aligned} \frac{d}{dt} \|v_-(t)\|_{L^2(\Omega)}^2 &= \frac{d}{dt} \int_{\Omega} v_-^2(t) \\ &= 2 \int_{\Omega} \frac{\partial v_-}{\partial t}(t) v_-(t) \\ &= 2 \int_{\Omega} \frac{\partial v}{\partial t}(t) v_-(t) \\ &= -2D(\nabla v(t), \nabla v_-(t))_{\Omega} - 2(f(u(t), u_b(t), v(t)), v_-(t))_{\Gamma} + (r(v(t)), v_-(t))_{\Gamma} - d_v(v(t), v_-(t))_{\Omega}, \\ &= -2D\|\nabla v_-(t)\|_{L^2(\Omega)}^2 - 2(f(u(t), u_b(t), v(t)), v_-(t))_{\Gamma} + (r(v(t)), v_-(t))_{\Gamma} - d_v\|v_-(t)\|_{L^2(\Omega)}^2, \end{aligned}$$

□

This result allows us to define another operator $\mathcal{T}_2 : \mathcal{X}^2 \rightarrow \mathcal{Y}$ sending a tuple $(u^*, u_b^*) \in \mathcal{X}^2$ to a solution $v \in \mathcal{Y}$ of Equation (3.8). By combining this step with the previous one, we can use $\mathcal{T} := \mathcal{T}_2 \circ \mathcal{T}_1 : \mathcal{Y} \rightarrow \mathcal{Y}$ to iteratively get new solutions. At each step, we modify Equation (3.7) using the input from $\mathcal{Y}$ to get a new pair $(u, u_b) \in \mathcal{X}^2$, with which we proceed to update Equation (3.8) to get a new $v \in \mathcal{Y}$.

Lemma 3.5

For $\tilde{T} > 0$ small enough, $\mathcal{T} : \mathcal{Y} \rightarrow \mathcal{Y}$ is a contraction. That is, there is a $q \in [0, 1)$ such that

$$\|\mathcal{T}v\|_{\mathcal{Y}} \leq q \|v\|_{\mathcal{Y}}$$

for every $v \in \mathcal{Y}$.

Proof.

□

proof

We have now gathered enough results to prove the main statement of this section.

Proof of Theorem 3.2. Using Lemma 3.5, let $T > 0$ such that $\mathcal{T}$ is a contraction. Then by the Banach fixed-point theorem, there exists a unique fixed point v^* of $\mathcal{T}$. Next, define $(u^*, u_b^*) := \mathcal{T}_1 v^*$. Then by definition, u^* and u_b^* solve Equation (3.7) with v^* as the frozen input. On the other hand, $\mathcal{T}_2(u^*, u_b^*) = \mathcal{T}v^* = v^*$ solves Equation (3.8) with u^* and u_b^* as the inputs to the frozen reactions. In other words, u^* , u_b^* and v^* are weak solutions to Equations (3.1) to (3.3).

To show uniqueness of the weak solutions, it is enough to note that for any weak solution u^* , u_b^* and v^* , we have that v^* is a fixed point of $\mathcal{T}$. This follows directly from the uniqueness results from Lemmas 3.3 and 3.4. Since the Banach fixed-point theorem tells us that fixed points are unique, we know that weak solutions to Equations (3.1) to (3.3) are too. □

Chapter 4

Numerical Analysis

We have seen several qualitative properties of solutions to our model. Some properties, however, are more difficult to discuss using purely analytical methods. One of these properties is the ability to generate Turing patterns. To investigate this, we turn to numerical methods and simulate the model to have a more concrete and clear view of its behaviour.

4.1 An expanded model

Now that we are working with simulations, we can make our model slightly more complex by dropping one of our assumptions. Let Ω and Γ be constructed as in section 3.1. In this chapter, we restrict our study to the dimensions of $d = 2$ and $d = 3$, since these are the most relevant to the biological applications of this model. Recall that in section 3.1, we assumed the receptors, both free and bounded, to be fixed in place on the cell membrane. This resulted in a system of one PDE and two ODEs, the latter being both defined separately for every point on the cell membrane. We now drop this assumption by adding diffusion terms to both of these equations, forming a new system of reaction diffusion PDEs given by

$$\begin{aligned}
 u_t &= D_u \Delta u + P_u + \alpha u_b - \delta_u u - 2k_{\text{on}} u^2 v + 2k_{\text{off}} u_b & \text{on } (0, T) \times \Gamma, \\
 u_{b,t} &= D_{u_b} \Delta u_b - \delta_{u_b} u_b + k_{\text{on}} u^2 v - k_{\text{off}} u_b & \text{on } (0, T) \times \Gamma, \\
 v_t &= D_v \Delta v - \delta_v v & \text{in } (0, T) \times \Omega, \\
 D_v \frac{\partial v}{\partial \nu} &= P_v - k_{\text{on}} u^2 v + k_{\text{off}} u_b & \text{on } (0, T) \times \Gamma, \\
 D_v \frac{\partial v}{\partial \nu} &= 0 & \text{on } (0, T) \times \Gamma,
 \end{aligned} \tag{4.1}$$

where D_u and D_{u_b} are both new positive real coefficients. We also have some corresponding initial conditions

$$u(0, x) = u_0(x), \quad u_b(0, x) = u_{b,0}(x) \quad \text{and} \quad v(0, y) = v_0(y)$$

for all $x \in \Gamma$ and $y \in \overline{\Omega}$.

There are multiple things to note about these new equations. Since Γ has an empty interior, it is impossible to define spatial derivatives for functions living exclusively on this set. This means we would not be able to use the Laplace operator to include diffusion of the receptors as we do in Equation (4.1). This issue is solved by using something called the Laplace-Beltrami operator instead, which is a generalisation of the Laplace operator to Riemannian manifolds. Since we assumed Γ to be a C^1 boundary, it forms such a manifold. The Laplace-Beltrami operator is further discussed in [Jos17], but we will not go into detail here. For our purposes, it is sufficient to note that it behaves in a similarly to the classical Laplace operator and admits analogous computational identities such as partial integration. It is sometimes denoted by Δ_Γ , but for the sake of simplicity, we will omit this notation and use Δ for both the Laplace and Laplace-Beltrami operators.

The space variable is now more involved in the equations for u and u_b , possibly making it seem necessary to define some boundary conditions. However, recalling our exact construction of the domain, we can see that the set Γ representing the cell membranes, when viewed as a separate $(d - 1)$ -dimensional manifold, does not have a boundary. Since the equations for u and u_b strictly live on this manifold, boundary conditions are not required.

4.2 The finite element method

In any course on elementary numerical methods, the simulation of ODEs will inevitably be discussed. The vast majority of these simulations include defining a small time step using the ODE to derive a suitable iteration. In the study of PDEs, one might think to extend this approach to the spacial variable. This would result in defining a time step $\tau > 0$ as well as a space step $h > 0$ and discretising the domain $(0, T) \times \Omega$ as a subset of $\tau\mathbb{Z} \times h\mathbb{Z}^d$. This is called the finite *difference* method, and it is very successful in a number of applications. However, this method is fairly hard to use when the dimension gets higher than $d = 1$ or $d = 2$, or the shape of the domain becomes more complex than, for instance, a rectangle. Our model lives on a domain that contains holes that are designed to model cells and as a result, will contain some curvature. The finite difference method is too unreliable in this case, so we need a more robust and adaptable approach.

The finite *element* method is another means to approximate solutions to a PDE. It is mainly used for discretising the space variable, so let us focus on an example that does not include time. We consider the standard Poisson equation

$$\Delta u = f \quad \text{in } \Omega \tag{4.2}$$

with Neumann boundary conditions

$$\frac{\partial u}{\partial \nu} = g \quad \text{on } \partial\Omega. \tag{4.3}$$

Here, $\Omega \subseteq \mathbb{R}^d$ is an open and bounded set with C^1 boundary and f and g are known functions on Ω and $\partial\Omega$, respectively. To discretise this spatial domain, we use the same approach as in [DE13] to define a *mesh*. We choose a finite number of points $\{x_i\}_{i=1}^m$ in Ω and use them as endpoints of d -simplices¹, which we collect in a set $\mathcal{T}_h$. These points are called *nodes*, and the d -simplices they

¹An n -simplex is the n -dimensional equivalent of a triangle. The most important instances are the 0-simplex, which is a point, the 1-simplex, a line segment, the 2-simplex, a triangle, and the 3-simplex, a tetrahedron. Every n -simplex is enclosed by $n + 1$ $(n - 1)$ -simplices, which in turn are enclosed by $(n - 2)$ -simplices, etc. All of these are referred to as its subsimplices.

enclose are referred to as *elements*. To have a well-behaved discretisation, for any two d -simplices $T, T' \in \mathcal{T}_h$ we require $T \cap T'$ to either be empty or form a common subsimplex of dimension $k < d$.

This leaves us with an approximation of Ω , denoted by

$$\Omega_h := \bigcup_{T \in \mathcal{T}_h} T.$$

The subscript h derives from the idea that this still represents the fineness of our discretisation, just as in the finite difference method. If we define

$$h := \max_{T \in \mathcal{T}_h} \text{diam}(T),$$

we want Ω_h to approximate Ω increasingly well as h approaches 0.

Additionally, we want a discretised function space, from which we look for approximate solutions to Equations (4.2) and (4.3), called the *trial space*. To this end, for every node x_i , we define a basis function $\phi_i : \Omega_h \rightarrow \mathbb{R}$ such that $\phi_i(x_i) = 1$, $\phi_i(x_j) = 0$ for all $j \neq i$ and ϕ_i is linearly interpolated on each element. Now the function space we consider for our trial functions is the space of linear combinations of these basis functions,

$$\mathcal{V}_h := \text{span}\{\phi_1, \dots, \phi_m\}.$$

This is called the space of *first-order Lagrange elements*, derived from the fact that the functions are piecewise linear polynomials on each element. There are many more kinds of function spaces used in finite element simulations, such as second-order Lagrange spaces, where the functions are piecewise quadratic polynomials on each element, or spaces where the functions are allowed to be discontinuous, referred to as discontinuous Galerkin elements.

We can now use this setup to convert our original problem into one that a computer can handle. Let $u_h \in \mathcal{V}_h$ be an approximate solution to Equations (4.2) and (4.3). Then for any $i \in \{1, \dots, m\}$, we can multiply the Poisson equation by ϕ_i , integrate on Ω_h and use partial integration to get

$$\int_{\Omega_h} f \phi_i = \int_{\Omega_h} \Delta u_h \phi_i = \int_{\partial\Omega_h} \frac{\partial u_h}{\partial \nu} \phi_i \, dS - \int_{\Omega_h} \nabla u_h \cdot \nabla \phi_i = \int_{\partial\Omega_h} g \phi_i \, dS - \int_{\Omega_h} \nabla u_h \cdot \nabla \phi_i.$$

Rearranging terms and writing u_h as its linear decomposition with coefficients $U_1, \dots, U_m \in \mathbb{R}$, we derive

$$\int_{\partial\Omega_h} g \phi_i \, dS - \int_{\Omega_h} f \phi_i = \int_{\Omega_h} \nabla \left(\sum_{j=1}^m U_j \phi_j \right) \cdot \nabla \phi_i = \sum_{j=1}^m U_j \int_{\Omega_h} \nabla \phi_j \cdot \nabla \phi_i.$$

If we define

$$F_i := \int_{\partial\Omega_h} g \phi_i \, dS - \int_{\Omega_h} f \phi_i \quad \text{and} \quad K_{ij} := \int_{\Omega_h} \nabla \phi_j \cdot \nabla \phi_i$$

for every $i, j \in \{1, \dots, m\}$, this is equivalent to the linear system

$$KU = F. \tag{4.4}$$

ugly??

example

example

Here, K is typically referred to as the *stiffness matrix*. We have now transformed the PDE into a linear system of equations, which we can solve using a wide variety of algorithms.

Before deriving the linear system for Equation (4.1), we should quickly illustrate how time discretisation ties into this approach. Consider, therefore, the heat equation

$$u_t = \Delta u + f \quad \text{in } (0, T) \times \Omega \quad (4.5)$$

with Neumann boundary conditions

$$\frac{\partial u}{\partial \nu} = g \quad \text{on } [0, T) \times \partial\Omega, \quad (4.6)$$

where $T > 0$ is some final time and now f and g are functions on $(0, T) \times \Omega$ and $[0, T) \times \partial\Omega$, respectively. Of course, we also need initial conditions

$$u(0, x) = u_0(x). \quad (4.7)$$

Similarly to the numerical study of ODEs, we choose a number of steps $N \in \mathbb{N}$ and let $\tau := T/N$. For every $n \in \{0, \dots, N\}$, we seek to approximate the time step

$$u_n(x) := u(n\tau, x).$$

To this end, we use a difference quotient in time to derive

$$\frac{u_{n+1}(x) - u_n(x)}{\tau} \approx \Delta u_{n+1}(x) + f(n\tau, x),$$

where we have used implicit diffusion and explicit forcing. This is a widely used scheme, commonly referred to as IMEX. Other schemes can be used with varying results in numerical stability and accuracy. In this thesis, we will not go too far into this choice.

After rearranging terms, this leads to the time-independent problem

$$u_{n+1}(x) \approx \tau \Delta u_{n+1}(x) + u_n(x) + \tau f(n\tau, x).$$

We can now use the same approach as before, transforming this problem into a linear system of equations

$$(M + \tau K)U^{n+1} = MU^n + \tau F^n,$$

where $U^n \in \mathbb{R}^m$ contains the coefficients for the linear decomposition of u_n into $\phi_1, \dots, \phi_m$, and F^n and M are defined by

$$F_i^n := \int_{\partial\Omega_h} g(n\tau, x) \phi_i(x) dS(x) + \int_{\Omega_h} f(n\tau, x) \phi_i(x) dx \quad \text{and} \quad M_{ij} := \int_{\Omega_h} \phi_j \phi_i$$

for $i, j \in \{1, \dots, m\}$. M is often called the *mass matrix*. When calculating a new iteration U^{n+1} , we can assume the previous iterations are known, starting off with the initial conditions

$$U_i^0 := u_0(x_i), \quad i = 0, \dots, m.$$

Chapter 5

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